

FLOOR PLANNING USING AI

PROJECT REPORT

FOR

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PROBLEM STATEMENT

Propose a solution using text-to-image GenAI with respect to manual floor planning that yields limited design diversity and compromises structural accuracy.

Our project aims to develop an AI-driven floor planning model that can generate architectural layouts while referencing existing designs or user-provided images. Traditional AI-based solutions often lack adaptability to specific styles, making them less practical for architects and designers. By leveraging Generative AI techniques like Stable Diffusion and ControlNet, our model can intelligently interpret and transform input references into optimized, functional, and aesthetically coherent floor plans, enhancing both creativity and efficiency in architectural design.

OBJECTIVES

Floor planning is crucial in architectural design, ensuring that spaces are arranged to meet structural, functional, and aesthetic objectives. A well-designed floor plan maximizes available space, particularly in densely populated areas where land is limited. While traditional methods are effective, they are labor-intensive, time-consuming, and prone to human error, lacking flexibility and real-time adaptability.

As architectural projects grow in complexity, the need for intelligent, efficient solutions becomes more evident. AI bridges this gap by automating design processes, understanding spatial and functional constraints, and enhancing both accessibility and accuracy, paving the way for architectural innovation.

The objective of our project is to develop an **AI-driven floor planning system** that can generate diverse, high-quality architectural layouts from **text-based user inputs**. By leveraging **multiple text-to-image AI models** (Stable Diffusion, ControlNet, ZB-Tech, Cloudiq, RunwayML), our goal is to:

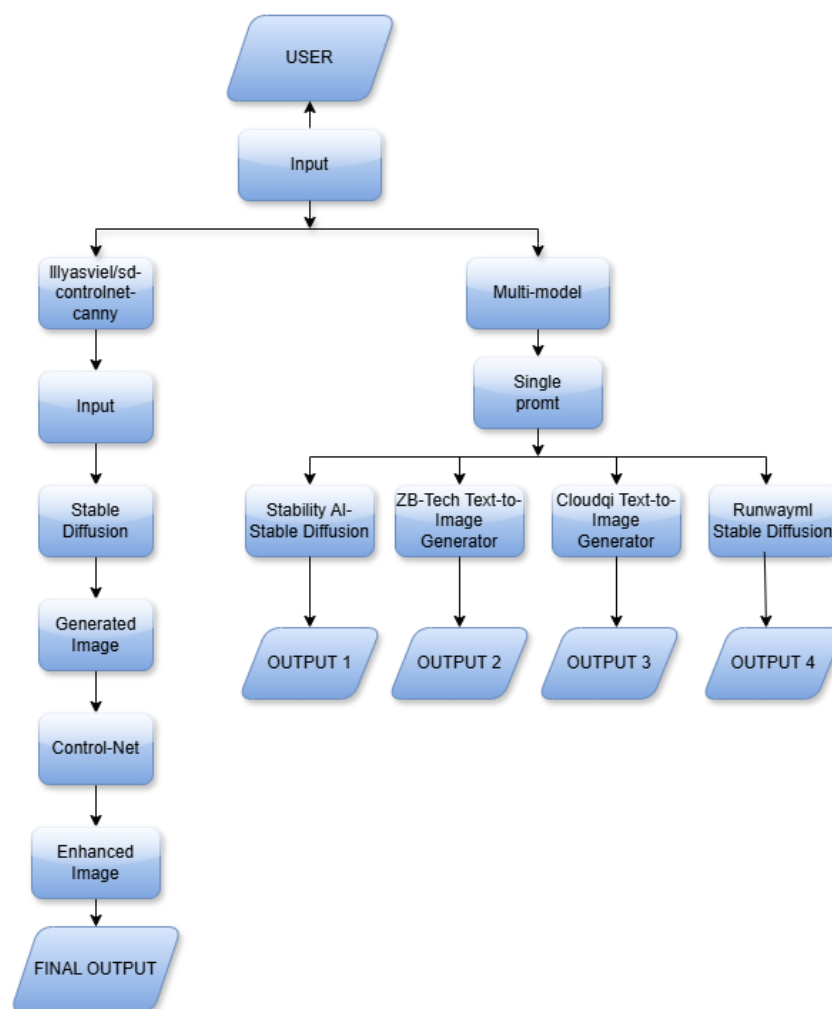
- **Automate the floor plan generation process**, reducing manual effort for architects.
- **Provide multiple design variations** from a single text prompt, enhancing creativity and decision-making.
- **Ensure structural accuracy and feasibility** using **ControlNet refinement**.
- **Make architectural planning more accessible and efficient** through AI-powered design generation.

METHODOLOGY

We understand that architects need either a specific floor plan or a diverse collection to choose from. To address this, we developed a platform that generates multiple designs from a single prompt using a multimodal approach, leveraging multiple AI models simultaneously. This ensures a wide range of floor plans, providing architects with valuable insights to refine their own designs. Additionally, our refinement process with ControlNet enhances structural accuracy and detail, ensuring the final output is realistic, optimized, and ready for implementation.

After careful observation and trial and errors we have selected the following technologies for our base model:

- Stable Diffusion: capable of generating architectural layout
- Control-Net: refine generated images by adding sharpness to the images
- Stability AI-Stable diffusion: suitable for detailed and high-quality outputs
- ZB-Tech Text-to-Image Generator: provides alternative design perspectives
- Cloudiq Text-to-Image Generator: adds more diversity in the architectural outputs
- Runwayml Stable Diffusion: ease of use and enhanced fine tuning capabilities



This flow chart should help to understand the work flow of the entire model structure that we have created. The process begins with a user interface where inputs are provided. Users can enter prompts to guide the AI in generating floor plans.

The system utilizes two major branches:
(a) Illyasviel/sd-controlnet-canny for structured image processing
(b) Multi-Model Text-to-Image Generators to generate various design outputs from a single prompt

In the first branch, Stable Diffusion processes input images to generate an initial design. The generated floor plan undergoes an additional refinement step using ControlNet, ensuring better alignment with architectural constraints, resulting in a final enhanced floor plan output.

In the second branch, different AI models generate diverse outputs based on the provided prompt.

Our model generates diverse floor plans from text-based inputs using a multi model approach, processing a single prompt through Stability AI, ZB-Tech, Cloudqi, and RunwayML for varied designs. This provides architects with multiple options for better decision-making. ControlNet then refines the images, enhancing structural accuracy and ensuring architectural feasibility, delivering an optimized floor plan ready for use.

To achieve this, we utilize Stable Diffusion and ControlNet for structured image generation, along with multiple text-to-image models for diversity. The backend is built using Python and Flask, while Google Colab is used for training and execution. Additionally, we integrate Hugging Face and OpenAI APIs for seamless AI model deployment.

DATASET AND TRAINING

Our model is trained on a diverse dataset of architectural floor plans from public and custom sources, including annotated residential layouts.

Following are dataset links and amount of dataset:

- Robin dataset (520)
- LIFULL HOME'S Dataset (80000)

We fine-tune pre-trained text-to-image models like Stable Diffusion using our dataset, employing supervised learning to understand spatial relationships in floor plans. ControlNet enhances structural accuracy, and training is conducted on Google Colab GPUs using Flask for optimization.

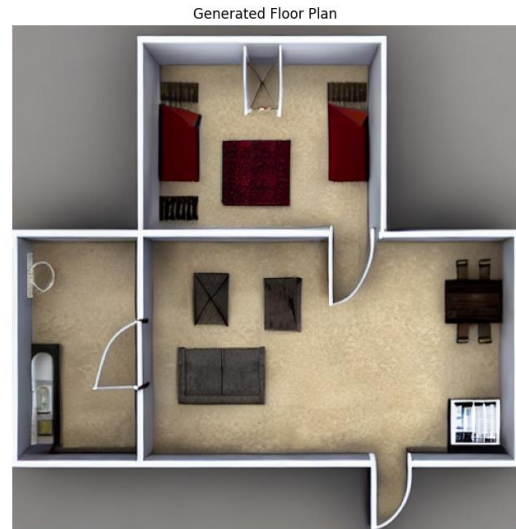
TRAILS AND ERRORS

To select the best base model for our AI, we went through many models, most of them failed to meet our requirements. The following table summarises the same.

Sl. No.	Model Name	Reason of failure
1	Janus (1.3 billion parameters)	Lack of resources to run
2	Janus pro (7 billion parameters)	Lack of resources to run
3	Vertex AI of Google	Lack of resources to run
4	Blip2	Text-to-Image reasoning, not a Text-to-Image Generator
5	Control-Net (Individually)	Failed to improve the image given
6	Control-Net (With diffusion model)	Fails on complex images.
7	HouseGAN (Pre-trained)	Fails to generate a meaningful image
8	HouseGAN (Un-trained)	Model requires specific dataset design solely for HouseGAN and we need to contact the owner for it.
9	Vison transform	Lack of adequate amount of dataset
10	GAN	Failed to generate a meaningful image

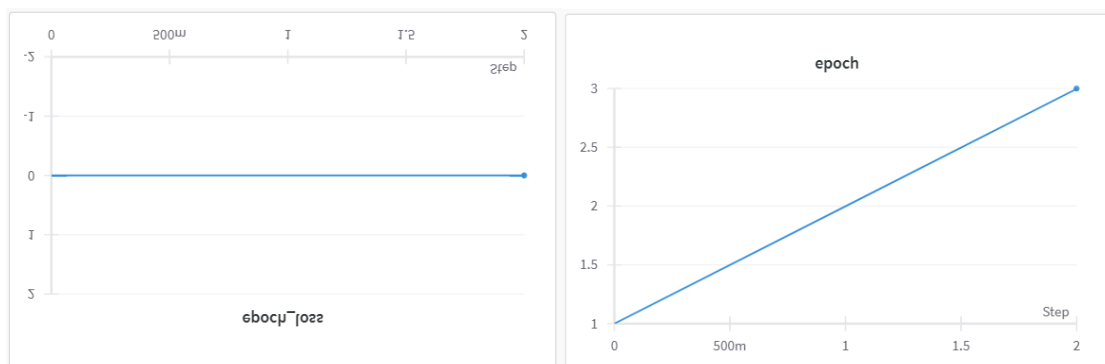
RESULTS

Despite the lack of a comprehensive dataset, our model delivered impressive results, generating diverse and structurally coherent floor plans. Leveraging multi-model AI processing and ControlNet refinement, we achieved high-quality outputs that align well with architectural standards. While a more extensive dataset could further enhance accuracy, the current results demonstrate the model's strong adaptability and potential for real world applications. Following are some the results generated by our model.





We also have two training graphs of our model:



These graphs were obtained using weights and biases

$$f(x) = x \cdot \sigma(x) = \frac{x}{1 + e^{-x}}$$

Activation function

CHALLENGES

During the development of our Floor Planning AI, we encountered multiple challenges that significantly impacted our model selection and training process.

- **Resource Constraints:** Many state-of-the-art models, such as Janus, Janus Pro, and Google's Vertex AI, required high computational power, which we lacked. This limited our ability to experiment with larger and more powerful models.
- **Dataset Limitations:** Finding a well-structured dataset for floor plans was a major hurdle. While searching across platforms like Kaggle, Hugging Face, and GitHub, we found very few relevant datasets. One promising dataset, LIFULL, contained Japanese floor plans, but the labels were in Japanese, making it difficult to integrate without extensive preprocessing and translation.
- **Model-Specific Requirements:** Some models, such as HouseGAN, required highly specific datasets designed for their architecture. The pre-trained version failed to generate meaningful images, while the untrained version demanded a custom dataset, for which we needed special access from the model creators.
- **Model Suitability:** Certain models, like BLIP-2, were designed for text-to-image reasoning rather than generation. Similarly, ControlNet worked well with basic image transformations but struggled with complex floor plan designs, even when combined with diffusion models.
- **Training Data Requirements:** Vision Transformers, which could have been a promising approach, required a large, high-quality dataset to perform effectively. However, due to the scarcity of labeled floor plan datasets, training such models became impractical.

These challenges led us to carefully evaluate alternative models and explore dataset augmentation techniques to improve our results.

FUTURE SCOPE

The future scope of the AI in the field of Floor planning are:

- **Real-Time Interactive Design & Customization** – AI can evolve to offer real-time modifications, allowing users to interactively adjust layouts and receive instant AI-generated recommendations based on preferences and constraints.
- **Integration with AR/VR for Immersive Visualization** – AI-generated floor plans can be integrated with **Augmented Reality (AR)** and **Virtual Reality (VR)** to provide immersive walkthroughs, helping architects and clients visualize spaces before construction.
- **AI-Driven Structural & Energy Optimization** – Future AI models can incorporate **structural stability analysis** and **energy efficiency predictions**, ensuring that floor plans are not only visually appealing but also optimized for sustainability and cost-effectiveness.

CONCLUSION

AI-driven floor planning represents a groundbreaking shift in architectural design, enabling faster, more efficient, and highly customizable solutions. By leveraging Generative AI models like Stable Diffusion and ControlNet, our approach enhances automation while preserving creative and functional integrity. This innovation not only streamlines the design process but also opens new possibilities for real-time customization, immersive visualization, and sustainable architecture. As AI continues to advance, its role in floor planning will expand, making architectural design more accessible, intelligent, and adaptable to diverse user needs.
